

List of Objects

Project Title: Coffee POS System

Team Name: POS

Team Members' Names:

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I. Entity Objects

1. **User** - Represents a system user who can access the Coffee POS system.

- userIdentifier, usernameCredential, encryptedPasswordHash, userEmailAddress, userRoleType, accountCreationTimestamp, accountActiveStatus

2. **StaffAccount** - Represents a staff member's account information and role permissions.

- staffAccountIdentifier, staffFullName, staffRoleType, staffEmailAddress, staffAccountStatus, accountCreationTimestamp

3. **Order** - Represents a customer order created and processed by a staff member.

- orderIdentifier, orderStatusType, orderSubtotalAmount, orderTaxAmount, orderTotalAmount, orderCreationTimestamp, createdByStaffIdentifier

4. **OrderItem** - Represents an individual menu item included in a customer order.

- orderItemIdentifier, associatedOrderIdentifier, associatedMenuItemIdentifier, orderedItemQuantity, itemCustomizationDescription, itemPriceAmount

5. **Payment** - Represents a payment transaction associated with a completed order.

- paymentTransactionIdentifier, associatedOrderIdentifier, paymentMethodType, paymentAmountValue, paymentCompletionStatus, paymentProcessingTimestamp

6. **Refund** - Represents a refund issued for a previously completed payment.

- refundTransactionIdentifier, associatedPaymentIdentifier, refundAmountValue, refundTypeCategory, refundProcessingTimestamp

7. **TransactionLog** - Stores records of financial transactions and system events.

- transactionLogIdentifier, associatedOrderIdentifier, transactionTypeCategory, transactionTimestamp, performedByUserIdentifier

8. **MenuItem** - Represents a beverage or food product offered for sale in the Coffee POS system.

- menuItemIdentifier, menuItemName, menuItemCategoryType, menuItemDescriptionText, menuItemPriceAmount, menuItemAvailabilityStatus

9. **InventoryItem** - Represents an ingredient or supply item used in preparing menu products.

- inventoryItemIdentifier, inventoryItemName, availableQuantityValue, minimumStockThreshold, inventoryLastUpdatedTimestamp

10. **QueueEntry** - Represents an order placed in the active order queue awaiting preparation.

- queueEntryIdentifier, associatedOrderIdentifier, queueEntryStatusType, queueEntryTimestamp

11. **SalesReport** - Represents a generated report summarizing sales performance.

- reportIdentifier, reportGenerationType, reportStartDate, reportEndDate, totalRevenueAmount, reportGeneratedTimestamp

II. Boundary Objects

1. **Login Boundary** - Interface where staff and administrators enter credentials to log into the POS system.
2. **Staff Accounts Boundary** - Administrative interface for creating, modifying, or deactivating staff accounts.
3. **New Order Boundary** - Interface used by staff to create new customer orders.
4. **Modify Order Boundary** - Interface that allows staff to update an existing order before payment.
5. **Checkout Boundary** - Interface where staff process payments for customer orders.
6. **Refund Boundary** - Interface used by administrators to issue refunds.
7. **Transaction Logs Boundary** - Interface displaying historical transaction records.
8. **Owner Menu Boundary** - Admin interface for managing menu items.
9. **Inventory Boundary** - Interface for updating stock levels.
10. **Staff Menu Boundary** - Interface where staff view available menu items.
11. **Active Orders Queue Boundary** - Interface displaying all pending and in-progress orders.
12. **Reports Boundary** - Interface displaying sales analytics and performance metrics.

13. **Export Reports Boundary** - Interface used to export reports into CSV or PDF formats.

III. Controller Objects

1. **AuthController** - Validates login credentials and manages authenticated user sessions.
2. **UserController** - Handles creation, modification, and deletion of staff accounts.
3. **OrderController** - Manages creation and modification of customer orders.
4. **OrderStatusController** - Updates order status values such as Pending, In Progress, and Completed.
5. **PaymentController** - Handles payment processing and stores payment records.
6. **RefundController** - Processes refund requests and updates payment records.
7. **TransactionController** - Records financial activities and manages transaction logs.
8. **MenuController** - Handles menu item operations.
9. **InventoryController** - Handles stock updates and thresholds.
10. **QueueController** - Handles updates and synchronization of the active orders queue.
11. **ReportController** - Generates sales analytics and report data from transaction records.
12. **ExportController** - Handles exporting report data into downloadable files.